

Michael Jia

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Availability May 2026 – December 2026

EDUCATION

Northeastern University

Boston, MA

Bachelor of Science in Computer Science, Minor in Mathematics

Expected May 2028

- **GPA: 3.5/4.0** | Dean's List
- Relevant Coursework: Algorithms, Object-Oriented Design, Intro to Databases, Artificial Intelligence, Foundations of Cybersecurity, Foundations of Data Science, Linear Algebra, Probability and Statistics
- Activities: Society of Asian Scientists and Engineers, AINU, Club Squash

EXPERIENCE

Software Engineering Intern

June 2025 – Present

TurboTides, Inc.

Remote

- Modernized the company's license generation workflow by replacing a single-user C# desktop tool with a multi-user web application (FastAPI, PostgreSQL, vanilla JS), centralizing customer data previously tracked in spreadsheets.
- Designed a relational database schema (customers → computers → licenses → features) with foreign key constraints and JWT-authenticated admin CRUD endpoints.
- Reverse-engineered the legacy C# tool's HMAC-SHA512 signing logic to replicate byte-identical license output in Python, tracing undocumented string assembly and date formatting quirks.
- Implemented a nightly scheduler for proactive expiry reminder emails and deployed the full stack on AWS EC2 with Docker Compose.
- Wrote the first automated database tests within the existing regression testing framework, transitioning the team from manual UI/DBeaver verification to reproducible script-based validation.
- Created JavaScript test scripts integrated with the C++ backend via exposed scripting APIs, automating CRUD validation against a dedicated PostgreSQL test database.
- Traced C++ source code to identify method signatures and isolate a critical memory crash triggered from the JS test runner, collaborating with senior engineers to resolve the issue.
- Built a Python image preprocessing pipeline (NumPy, Pillow) to normalize and standardize datasets, streamlining data preparation for the machine learning team.

PROJECTS

RL Market Making Agent | *Python, OpenAI Gym, NumPy, Pandas, Matplotlib*

April 2026

- Built a simulated limit order book environment from scratch to train a reinforcement learning agent on optimal bid-ask pricing strategies.
- Designed an MDP formulation (state: inventory/mid-price/volatility/time; action: bid/ask offsets; reward: trade profit minus inventory cost) and implemented a Q-learning algorithm.
- Led development for a team of 3; evaluated Q-learning and DQN model performance against a random baseline using mean average equity over 1,000 episodes.

Personal Automation Agent | *Python, SQLite, Discord.py, Gemini API, AWS EC2*

March 2026

- Built a Discord bot deployed as a systemd service on AWS EC2 automating technical interview prep and academic task tracking with scheduled and on-demand workflows.
- Integrated Gemini API with retry logic and structured JSON extraction to parse syllabus PDFs, manage tasks in SQLite, and generate time-aware daily schedules.
- Automated a LeetCode study pipeline via GraphQL API, generating type-hinted Python starter files via LLM and committing to GitHub on a daily cron schedule.

TECHNICAL SKILLS

Languages: Java, Python, SQL (PostgreSQL, MySQL), JavaScript, HTML

Libraries & Frameworks: TensorFlow/Keras, NumPy, OpenCV, scikit-image, Pillow, FastAPI, Pydantic

Cloud & DevOps: AWS (EC2, RDS, Security Groups), Docker/Docker Compose, Git/GitHub

Tools: IntelliJ, Visual Studio, DBeaver, DataGrip, Jupyter Notebook